LumiFuse: Illumination-Adaptive Hybrid Low-Light Image Restoration Combining Retinex Priors with Zero-Reference Curve Estimation

Hackathon Problem P1 — Team T81 (Bugweiser)

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Abstract—Images captured in low light are dark, noisy and colour-distorted, and any gain that lifts the shadows also amplifies noise and clips regions that were already lit. We present LumiFuse, an illumination-adaptive restoration system that measures each image first and derives every parameter from those measurements. It runs two branches: an adaptive classical pipeline (guided-filter Retinex decomposition, gain-aware edge-preserving denoising, adaptive CLAHE and colour correction) and a 1.0 M-parameter network built on Zero-DCE whose curve estimator and refinement U-Net are conditioned on the classical illumination map and on a noise-amplification map. A reliability-weighted fusion combines the branches per pixel, and no-reference quality gates correct exposure, residual noise and colour cast. Trained on a de-duplicated, leak-free LOL split (we found that 89 of the 100 LOL-v2-real test images are LOL-v1 training images), LumiFuse reaches 18.09 dB PSNR / 0.7375 SSIM on LOL-v1 (+3.23 dB over official Zero-DCE), the best SSIM on four of five paired test sets, introduces no clipped pixels, and runs in 0.53 s per 600×400 frame on a laptop CPU. We report a 16-variant ablation, artefact metrics and an exposure-ambiguity failure analysis. Code, weights and evaluation: https://github.com/dhruvo11/lumifuse-lowlight; live demo: https://lumifuse.streamlit.app.

Index Terms—low-light image enhancement, Retinex, Zero-DCE, guided filter, CLAHE, hybrid restoration, image quality assessment

I. Introduction

Low-light photography suffers from five coupled degradations: severe under-exposure, sensor noise, colour casts, non-uniform illumination (lamps, windows, backlight) and loss of texture after denoising. Classical methods based on Retinex theory [1]–[3] are fast and interpretable but use hand-set parameters and amplify noise; learned enhancers [4]–[8] restore better on their training distribution but are opaque and can fail on unseen exposure levels. This work addresses the hackathon challenge of building an adaptive, deployable system that combines both and is evaluated honestly on unseen data. Our contributions are:

- An adaptive classical pipeline in which illumination smoothing, re-lighting exponent, denoising strength, CLAHE clip limit and colour-correction strength are all computed from measured image statistics.
- LumiNet, a Zero-DCE [7] curve estimator extended with a classical illumination-prior channel plus a refinement U-Net conditioned on a classical noise-amplification map, fine-tuned from the official weights.
- A hybrid decision layer: two-sided input-exposure conditioning, reliability-weighted per-pixel fusion with a brightness-dependent global trust term, and no-reference quality gates.
- A leak-free protocol (test-overlap removal), evaluation on six test sets with all metrics requested by the problem statement, a 16-variant ablation, artefact metrics, failure analysis, and a public web deployment.

II. Related Work

Retinex and classical enhancement. Retinex models an image as $I = R \odot L$ [1]; MSRCR [2] and LIME [3] estimate and adjust the illumination L, LIME refining a max-RGB initial map with a structure-aware prior. CLAHE [9] equalises local histograms with a clip limit; bilateral [10] and guided [11] filters smooth noise while keeping edges. Learned enhancement. RetinexNet [4] introduced the paired LOL dataset and a decomposition network; KinD [5] adds a dedicated denoising branch; EnlightenGAN [6] trains without pairs; Zero-DCE [7] and Zero-DCE++ [12] estimate pixel-wise quadratic curves with zero-reference losses; Retinexformer [8] is a recent transformer. LumiFuse keeps the efficiency and monotonicity of curve estimation and injects classical priors where curves are weakest: noise and colour.

III. Method

Fig. 1 shows the inference path; module and function names in the figure match the repository.

A. Image analysis

For each input we measure mean/median luminance, the fraction of dark ($Y < 0.1$) and bright ($Y > 0.85$) pixels, illumination non-uniformity (standard deviation of the log of a heavily blurred max-RGB map), colour cast

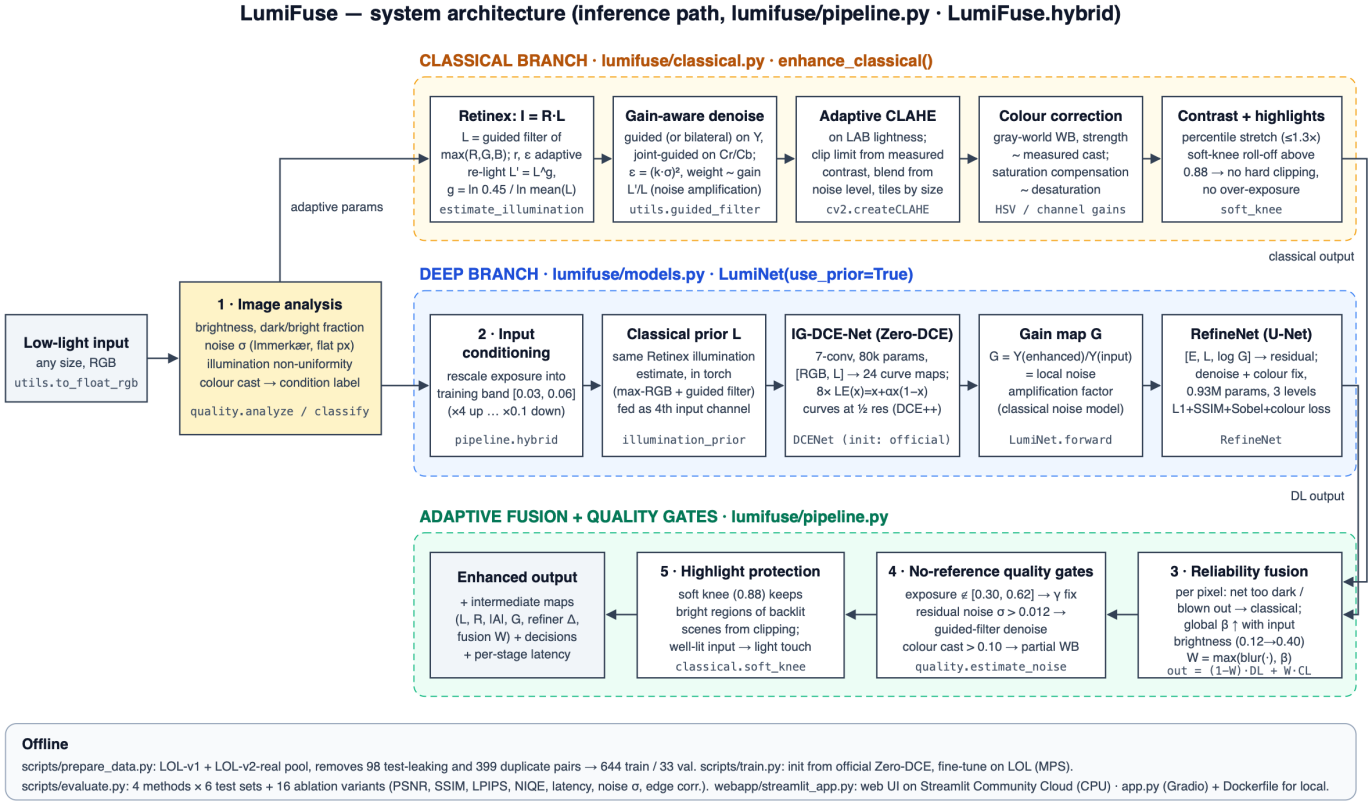


Fig. 1. LumiFuse architecture. An analysis stage measures the input; a classical branch (top) and an illumination-guided deep branch (middle) restore it in parallel; reliability fusion and no-reference quality gates (bottom) produce the output together with all intermediate maps.

$(\max_c \mu_c - \min_c \mu_c)/\bar{\mu}$, and the noise standard deviation σ with Immerkær's operator [13] restricted to the flattest 50 % of pixels so that texture is not counted as noise. These values select every parameter below and a condition label shown to the user.

B. Adaptive classical branch

Illumination and re-lighting. $L_0 = \max_c I_c$ is refined by a self-guided filter $L = \mathrm{GF}(L_0, L_0; r, \epsilon)$ with $r = \min(H, W)/50$ and $\epsilon = (0.1\, P_{90}(L_0))^2$, which makes the smoothing scale-invariant to exposure. The illumination is re-lit as $L' = L^g$ and the image as $\hat{I} = I \odot L'/L$, with

$$g = \mathrm{clip}\left(\frac{\ln 0.45}{\ln \exp(\overline{\ln L})},\, 0.15,\, 1 \right), \tag{1}$$

mapping the log-average illumination to the brightness of well-exposed references. Because $L'/L = L^{g-1}$, dark regions are lifted most and regions with $L \approx 1$ are left unchanged, which protects highlights. Gain-aware denoising. After re-lighting, σ is re-measured; luma is filtered by a guided filter with $\epsilon_Y = (5\sigma)^2$ (bilateral is selectable) and chroma by a joint guided filter guided by the clean luma. The filtered result is blended per pixel with weight $\propto (L'/L)$, so noise suppression follows amplification. CLAHE is applied to LAB lightness with clip limit $1 + 2.5(0.22 - c)/0.22 \in [1, 3]$ from the RMS contrast c and a lighter blend for noisy images. Colour correction applies gray-world gains with strength proportional to the measured cast, and saturation compensation proportional to the measured desaturation. A percentile contrast stretch (skipped when $\sigma > 0.02$) and a soft-knee roll-off above 0.88 complete the branch. With adaptive=False each rule is replaced by a fixed constant.

C. Illumination-guided deep branch (LumiNet)

Curve stage. The Zero-DCE estimator (seven 3×3 convolutions, 32 channels) receives $[I, L]$, where L is the same guided-filter illumination computed in PyTorch, and predicts eight three-channel curve maps A_n applied iteratively:

$$E_n = E_{n-1} + A_n \odot E_{n-1} \odot (1 - E_{n-1}), \quad E_0 = I. \tag{2}$$

Curves are estimated at half resolution and bilinearly up-sampled as in Zero-DCE++ [12]. The network is initialised from the authors' weights; the new input channel is zero-initialised, so training starts exactly at official Zero-DCE. Refinement. Curves can brighten but cannot denoise. A three-level U-Net [14] (0.93 M parameters) receives $[E, L, \frac{1}{4}\log G]$ with the amplification map $G = Y(E)/(Y(I) + 0.02)$—a classical noise model telling the network how much each pixel's noise was multiplied—and predicts a residual: $\hat{Y} = E + \mathcal{R}_\theta(E, L, G)$. Losses. The

refiner is trained with $\mathcal{L}_1 + 0.2(1-\mathrm{SSIM}) + 0.1\,\mathcal{L}_1(\nabla_S) + 0.1\,\mathcal{L}_\angle$, where ∇_S are Sobel gradients (edge preservation) and $\mathcal{L}_\angle$ is the cosine distance between RGB vectors (colour shifts). The curve stage is trained with a low-pass $\mathcal{L}_1$ (4×4 average pooling, since curves should not chase noise), $0.5\,\mathcal{L}_\angle$, and the Zero-DCE illumination-smoothness term $10\,\mathrm{TV}(A)$. The curve output is detached before the refiner; without this, joint optimisation collapsed the curve stage into a feature map (mean R/G outputs 0.07/0.92).

D. Hybrid decision layer

Input conditioning. The network learned the exposure mapping of LOL inputs (mean luminance $\mu_0 = 0.05$). Inputs outside $[0.6\mu_0, 1.2\mu_0]$ are rescaled by $k = \mu_0/\mu$ (at most $4\times$) or $k = 1.2\mu_0/\mu$ (at least $0.1\times$). Reliability fusion. With Y_D, Y_C the luminance of the deep and classical outputs, $u = \mathrm{clip}((0.12 - Y_D)/0.12)\,[Y_C > Y_D]$ marks regions left too dark and $o = \mathrm{clip}((Y_D - 0.9)/0.1)\,[Y_C < Y_D]$ regions blown out; a global trust term $\beta = \mathrm{clip}((\mu - 0.12)/(0.40 - 0.12), 0, 0.9)$ shifts weight to the classical branch for moderately dark inputs, where the network's strong denoising flattens contrast. Then

$$W = \max\big(\mathcal{G}_s(\max(u, o)),\, \beta\big), \qquad \hat{I} = (1-W)\odot D + W\odot C, \tag{3}$$

with a Gaussian blur $\mathcal{G}_s$, $s = \min(H, W)/80$, to avoid seams. Quality gates. If the mean output luminance leaves $[0.30, 0.62]$ a luminance-only gamma restores it; if residual $\sigma > 0.012$ a guided-filter denoise is applied; if the colour cast exceeds 0.10 a partial gray-world correction is applied; a soft knee protects highlights. Well-lit inputs ($\mu > 0.35$) are blended back towards the original.

IV. Experimental Setup

Data and leak-free split. We pooled LOL-v1 our485 [4] and LOL-v2-real Train [15]. Hashing every low-light input revealed that 89 of the 100 LOL-v2-real test images are pixel-identical to LOL-v1 training images. We removed all 98 pairs overlapping either test set and 399 duplicates, leaving 644 training and 33 validation pairs. The validation split—never the test sets—was used for checkpoint selection and every hyper-parameter, including synthetic variants of validation images for exposure thresholds. Test sets. LOL-v1 eval15; LOL-v2-real test (100); three synthetic conditions built from 30 unseen LOL-v2 references with a Poisson–Gaussian camera model [16] (linearise, scale exposure, shot + read noise, re-encode, 8-bit quantise): moderate ($\times0.08$–0.15), extreme-dark ($\times0.003$–0.006) and non-uniform (gradient + spotlight, $\times0.004$–0.6); and 74 real DICM [17] and LIME [3] photographs. Metrics (in the order of the problem statement): PSNR and SSIM [18] on RGB; LPIPS (AlexNet) [19]; NIQE [20] (MATLAB-equivalent implementation); CPU latency; residual noise σ; edge preservation (correlation of Sobel magnitudes with the reference). No ground-truth-mean brightness correction is applied to any score. Training. 4000 iterations, batch 8, 160^2 crops with flips/rotations and illumination augmentation (30 % darkened $\times0.3$–1, 15 % brightened $\times1$–2, 20 % multiplied by a random smooth field), AdamW with one-cycle schedule (4×10^{-4} refiner, 2×10^{-4} curve stage), Apple M1 GPU, about 40 minutes per model. The DL-only baseline is the same network and recipe without the prior channels.

V. Results

Benchmark comparison. Table I compares the four approaches. Our official Zero-DCE result on LOL-v1 (14.86 dB) reproduces the value reported in [7], validating the metric code. LumiFuse obtains the best PSNR and SSIM on LOL-v1 (18.09 dB / 0.7375; +3.23 dB over Zero-DCE, +1.03 dB over the classical pipeline, +0.86 dB over DL-only). On LOL-v2 it ties DL-only on SSIM but its PSNR is below the classical pipeline; Section VI shows that this is a global-exposure choice rather than restoration error. DL-only has the best LPIPS and NIQE: the classical priors push the refiner towards stronger denoising, which raises fidelity at some cost in texture—the perception–distortion trade-off [21]. Fig. 2 shows examples.

Generalisation across illumination. Table II and Figs. 3–4 evaluate unseen lighting conditions. On extremely dark inputs LumiFuse is best on PSNR and SSIM (16.01 dB / 0.501 versus 0.317 classical and 0.083 Zero-DCE). Under moderate and non-uniform light it has the best SSIM (0.799 and 0.796) while the classical pipeline or Zero-DCE keep slightly higher PSNR. On real photographs the classical and Zero-DCE outputs have marginally lower NIQE (3.80–3.86 vs. 4.13); NIQE rewards high-frequency texture, including residual noise.

Over-exposure, colour shift and detail loss. Table III quantifies the artefacts named in the challenge. LumiFuse introduces no clipped pixels on any test set (the curve formulation cannot exceed 1 and the soft knee rolls off highlights), whereas the classical stretch clips up to 0.11 % on LOL-v1. Its hue error is close to the best on LOL-v2 and much lower than Zero-DCE on extreme-dark inputs (5.8° vs. 29.2°), and it preserves the most edge structure there. ΔE also counts brightness and therefore mirrors the exposure differences discussed below.

Intermediate representations. Fig. 5 shows the maps the system exposes: the classical illumination prior, reflectance, curve strength, curve-stage output, amplification map G, refiner correction, classical output and fusion weights. The refiner correction is largest exactly where G is large, confirming that the network uses the noise-amplification prior.

Ablation. Table IV removes one component at a time. The classical priors add +1.00 dB on LOL-v1 and +1.14 dB on LOL-v2 (on the synthetic sets the prior-free network is slightly better, and it always has better LPIPS/NIQE). Removing the refiner leaves LOL-v1 PSNR unchanged but lowers SSIM by 0.023–0.049 and

TABLE I
Comparison of classical, deep-learning and hybrid approaches on the paired LOL benchmarks (best in bold). LOL-v2-real test images never appear in training. Latency: Apple M1 CPU, 4 threads, 600 × 400 input.

	LOL-v1 eval15 (15 images)				LOL-v2-real test (100 images)				
Method	PSNR↑	SSIM↑	LPIPS↓	NIQE↓	PSNR↑	SSIM↑	LPIPS↓	NIQE↓	ms/frame
Classical (adaptive)	17.06	0.6612	0.386	6.42	19.12	0.6890	0.399	6.58	64
Zero-DCE (official)	14.86	0.5627	0.335	7.65	18.06	0.5798	0.312	7.90	276
DL-only (LumiNet, no priors)	17.23	0.7344	0.270	3.86	16.14	0.7382	0.296	3.96	308
LumiFuse (hybrid)	18.09	0.7375	0.318	4.87	17.41	0.7378	0.354	4.72	529



Fig. 2. LOL-v1 examples. Classical output is grainy, official Zero-DCE is washed out with colour casts, DL-only is clean but flat; LumiFuse balances exposure, noise and colour.

TABLE II
Generalisation across illumination conditions on unseen scenes: synthetic moderate, extremely dark and non-uniform (gradient + spotlight) lighting built from 30 LOL-v2 test references with a Poisson–Gaussian camera model, and 74 real DICM/LIME photographs (no reference, NIQE).

	Moderate (×0.08–0.15)			Extreme-dark (×0.003–0.006)			Non-uniform			Real
Method	PSNR	SSIM	LPIPS	PSNR	SSIM	LPIPS	PSNR	SSIM	LPIPS	NIQE
Classical (adaptive)	17.87	0.762	0.254	14.99	0.317	0.908	16.88	0.748	0.266	3.86
Zero-DCE (official)	18.50	0.738	0.245	10.34	0.083	1.300	16.82	0.720	0.274	3.80
DL-only (LumiNet, no priors)	13.53	0.737	0.314	15.78	0.490	0.746	13.94	0.738	0.322	3.94
LumiFuse (hybrid)	17.10	0.799	0.262	16.01	0.501	0.763	16.73	0.796	0.252	4.13

triples residual noise. Fine-tuning adds +3.03 dB on LOL-v1 and +2.0 dB on extreme-dark but costs 2.6 dB on LOL-v2, whose references are darker. Input conditioning (+1.75 dB) and the global trust term (+1.09 dB, SSIM +0.054) matter most under non-uniform light; the per-pixel reliability map rarely fires on these sets and acts as a safety net for blown highlights in real photos. Quality gates add +0.30 dB on extreme-dark and +0.14 dB on LOL-v2.

Table V ablates the classical branch. Retinex re-lighting is essential (−7.8 dB without it); denoising adds 1.2–3.3 dB; guided filtering beats bilateral filtering; adaptive parameters beat fixed ones by +1.79 dB (LOL-v1), +1.48 dB (LOL-v2) and +2.20 dB (extreme-dark). CLAHE, required by the challenge, lowers PSNR against the low-contrast LOL references while increasing local contrast; we therefore use an adaptive clip limit and a noise-dependent blend.

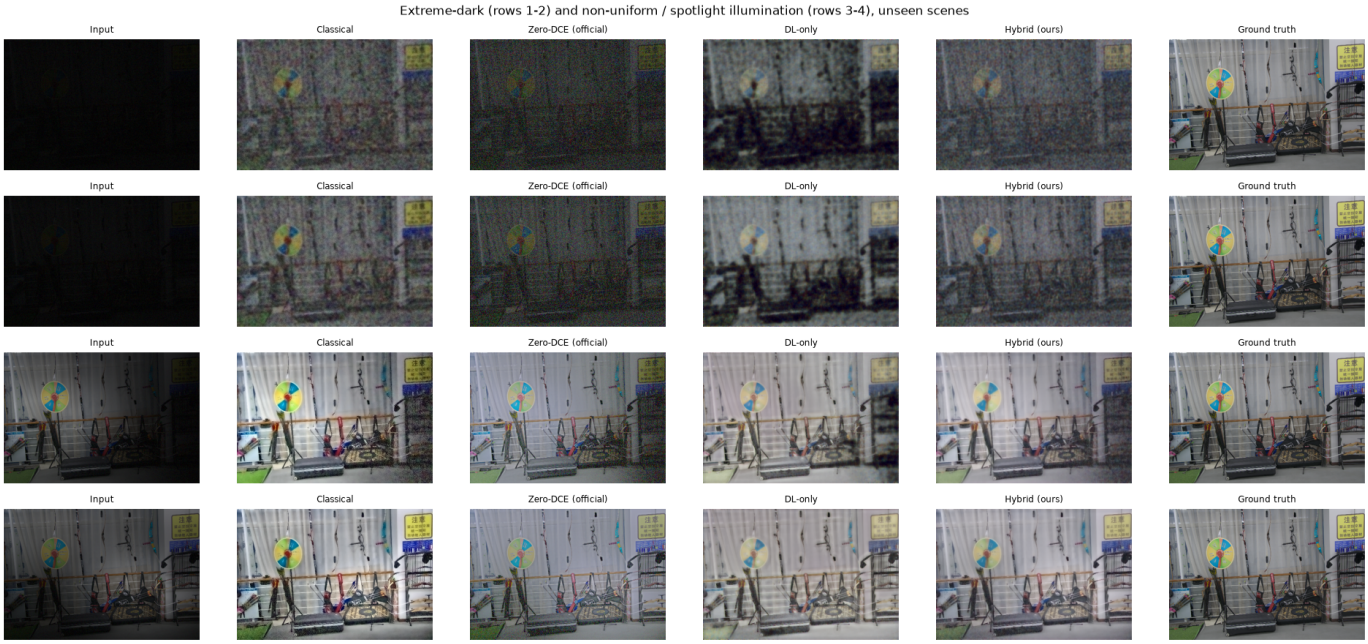


Fig. 3. Unseen scenes under extreme darkness (rows 1–2) and non-uniform spotlight illumination (rows 3–4).

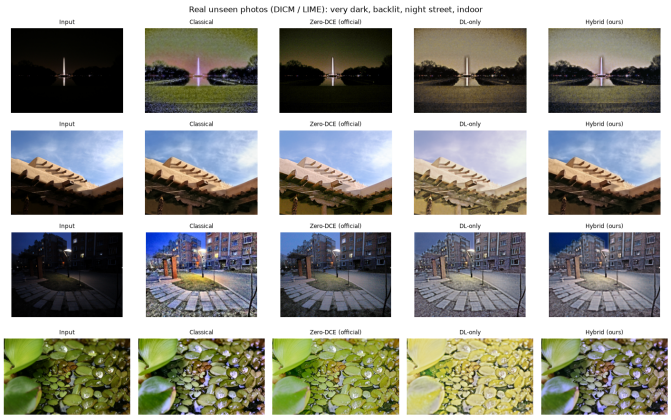


Fig. 4. Real photographs (DICM/LIME): very dark, backlit, night street and indoor scenes.

TABLE III
Artefacts: newly clipped pixels (%), colour shift (CIEDE2000 ΔE, RGB angle in degrees), edge preservation (Sobel-magnitude correlation) and residual noise σ.

Method	Clip%↓	$\Delta E\downarrow$	Angle↓	Edge↑	Noise $\sigma\downarrow$
LOL-v2-real test					
Classical	0.004	12.8	7.6	0.777	0.0151
Zero-DCE	0.000	14.2	12.4	0.882	0.0215
DL-only	0.000	15.4	4.1	0.839	0.0032
LumiFuse	0.000	13.5	4.3	0.792	0.0024
Synthetic extreme-dark					
Classical	0.000	16.0	4.2	0.192	0.0402
Zero-DCE	0.000	30.4	29.2	0.141	0.0910
DL-only	0.000	15.7	7.7	0.372	0.0088
LumiFuse	0.000	16.2	5.8	0.385	0.0074

VI. Failure Analysis and Limitations

Global exposure ambiguity. On a few LOL-v2 scene series (e.g. 00724–00730, 00763–00764) LumiFuse produces a mean brightness near 0.5 where the reference is near 0.33, losing up to ∼10 dB on those images (Fig. 6). Rescaling each output's mean brightness to the reference (a diagnostic that uses the ground truth and is never reported as a score) lifts LumiFuse to the best value on LOL-v2 (22.30 dB vs. 21.27 dB for the classical pipeline) and on every other paired set except moderate light, where it ties (Table VI). The deficit is therefore exposure choice, not restoration quality. Target brightness is ill-posed from a dark input: references use photographer-chosen exposures, the network learns their average, and PSNR penalises a global offset heavily (a uniform error of 0.1 alone caps PSNR at 20 dB). Using the classical illumination estimate as the exposure target lowered validation PSNR from 18.54 to 17.39 dB and was rejected. Mitigations: a user-controlled target exposure (Zero-DCE's exposure loss already takes a target) or exposure-conditioned training.

Extreme darkness. At ×0.003 exposure a white surface collects tens of photons and the shadows occupy 1–5 code values after 8-bit quantisation; every method has low SSIM there and LumiFuse leaves low-frequency chroma blotches. RAW or burst input and refiner training with synthetic Poisson–Gaussian noise are the natural fixes. Other limitations. The classical re-lighting exponent is global, so a bright spotlight reduces the lift of the dark side; strong point lights in real photographs show amplified lens glare; training data are mostly indoor LOL scenes.

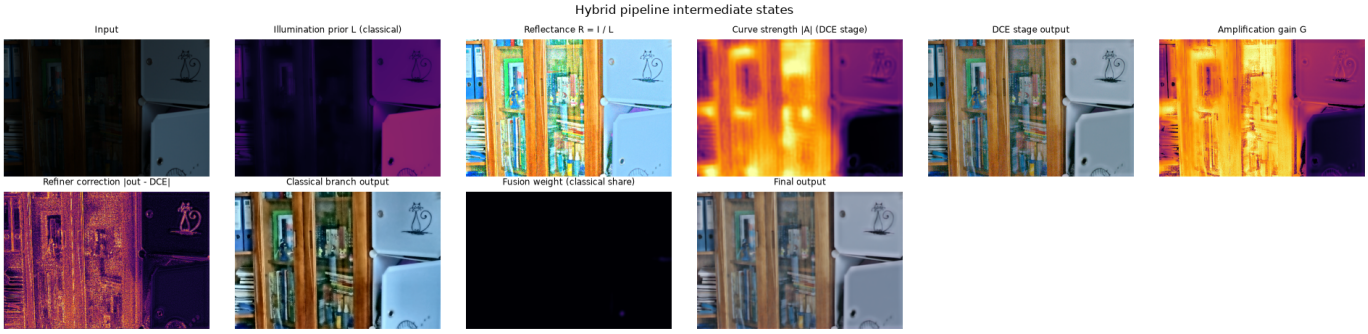


Fig. 5. Intermediate states of the hybrid pipeline for a LOL-v1 input.

TABLE IV
Ablation of the hybrid system: each row removes one component (PSNR / SSIM; best in bold).

	LOL-v1		LOL-v2-real		Extreme-dark		Non-uniform	
Variant	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
Full LumiFuse	18.09	0.737	17.41	0.738	16.01	0.501	16.73	0.796
− classical priors (L, G)	17.09	0.735	16.27	0.737	16.80	0.530	16.96	0.803
− RefineNet	18.08	0.714	16.85	0.725	17.33	0.513	16.25	0.748
− LOL fine-tuning	15.06	0.700	20.02	0.776	14.01	0.530	16.91	0.791
− input conditioning	18.10	0.736	17.24	0.735	15.95	0.501	14.98	0.762
− reliability fusion	18.09	0.737	17.40	0.737	16.01	0.501	15.64	0.742
− global trust β	18.09	0.737	17.40	0.737	16.01	0.501	15.64	0.742
− quality gates	18.13	0.738	17.27	0.738	15.71	0.501	16.73	0.796
− highlight protection	18.09	0.737	17.41	0.738	16.01	0.501	16.73	0.796

TABLE V
Ablation of the classical pipeline (PSNR / SSIM).

	LOL-v1		LOL-v2		Extreme	
Variant	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
Adaptive, full	17.06	0.661	19.12	0.689	14.99	0.317
Fixed parameters	15.27	0.639	17.64	0.639	12.79	0.115
− Retinex	9.24	0.345	11.40	0.355	7.55	0.150
− denoising	15.90	0.374	16.67	0.401	11.69	0.074
Bilateral (vs guided)	16.74	0.590	18.38	0.607	14.82	0.216
− CLAHE	17.17	0.707	20.24	0.736	13.25	0.466
− colour correction	17.11	0.665	18.47	0.689	15.16	0.319

TABLE VI
Exposure diagnostic: PSNR before and after rescaling each output's mean brightness to the reference (GT-mean; diagnostic only, never reported as a score). Bold: best GT-mean PSNR over all four methods. Last column: hybrid mean absolute brightness error.

	Classical		LumiFuse		
Set	PSNR	GT-mean	PSNR	GT-mean	$\lvert\Delta Y\rvert$
LOL-v1	17.06	19.65	18.09	20.98	0.081
LOL-v2	19.12	21.27	17.41	22.30	0.102
Moderate	17.87	20.06	17.10	20.06	0.094
Extreme	14.99	18.85	16.01	21.00	0.123
Non-unif.	16.88	18.23	16.73	18.94	0.078

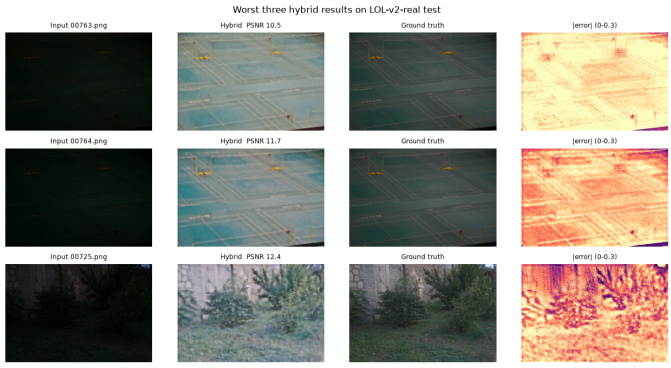


Fig. 6. Worst LumiFuse results on LOL-v2-real: output, reference and absolute error. Errors are dominated by global brightness.

Ethics. The refiner is a learned prior that can smooth or, rarely, invent fine texture, so outputs must not be used as forensic or medical evidence. The web application processes images in memory and stores nothing. Datasets are used under their research terms and are not redistributed.

VII. Deployment and Reproducibility

The system is deployed as a Streamlit web application on Streamlit Community Cloud (https://lumifuse.streamlit.app) with a before/after slider, intermediate maps, a compare-all mode, PNG download and explicit error states; inputs are capped at 768 px and a photo is processed in about 1.6 s on the cloud CPU. A Gradio version and a Dockerfile (built and run, 1.7 s per request in the container) provide one-command local use, and run_inference.py provides batch

inference. The repository (https://github.com/dhruvo11/lumifuse-lowlight) contains all source code, trained weights, the architecture diagram, a design document, the evaluation file eval_P1_T81.md, ablation and failure analyses and 23 robustness tests. A fresh clone that re-downloads all datasets and re-runs data preparation and evaluation reproduces the data split and all reported metrics to four decimal places.

VIII. Conclusion

LumiFuse shows that classical illumination analysis and a compact learned enhancer are complementary: classical priors improve the network's fidelity (+1.0–1.1 dB on real LOL data), measured image statistics make every stage adaptive, and reliability fusion with no-reference gates keeps the system safe on inputs outside the training distribution, all at about half a second per frame on a CPU. The main open problem is choosing the target exposure; exposure-conditioned training and a user brightness control are our next steps.

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